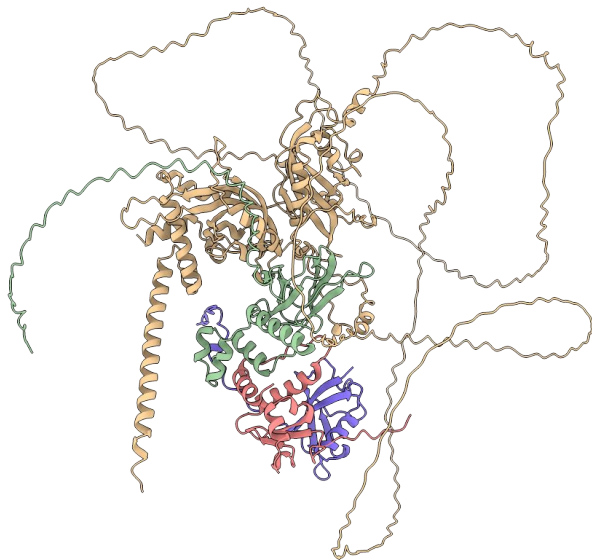




Package to assist in AlphaFold2¹ and AlphaFold3² related tasks

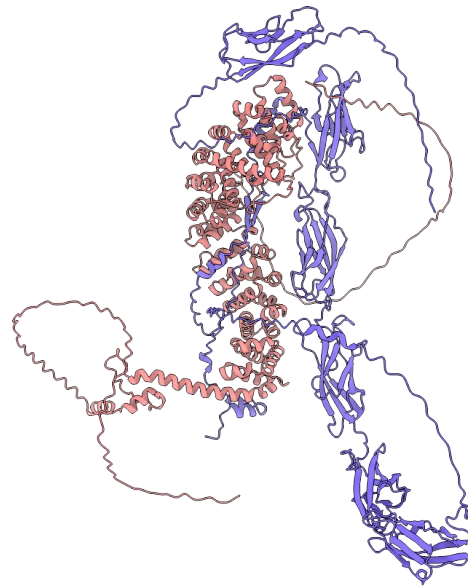
Why do we need AF-Pipeline?

HIF- α bound to pVHL/elongin-C/elongin-B complex



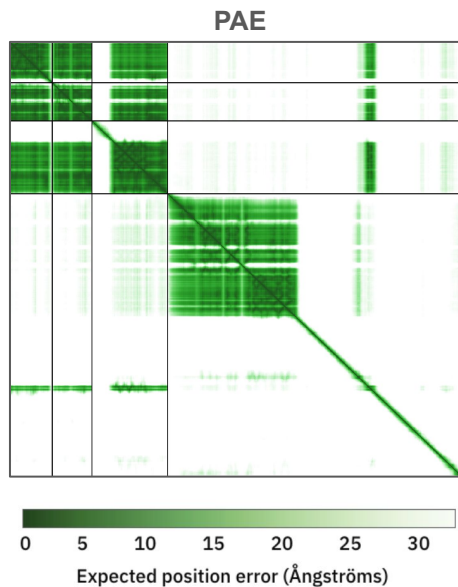
ipTM = 0.67
pTM = 0.4

Cadherin-catenin complex

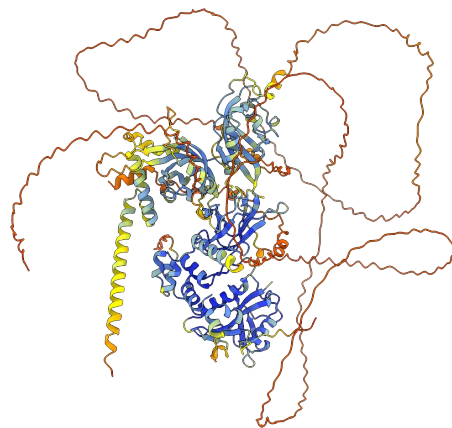


ipTM = 0.61
pTM = 0.46

Why do we need AF-Pipeline?



ipTM = 0.67
pTM = 0.4

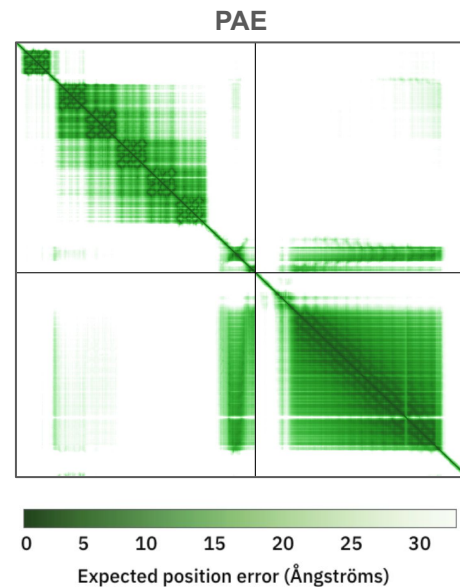


pLDDT score

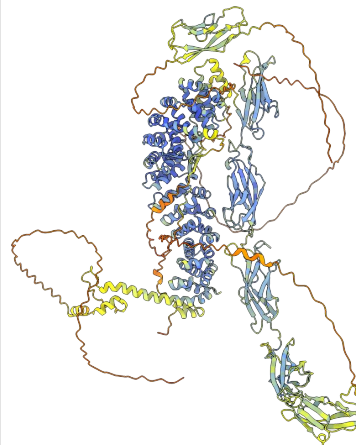
Score	Color
0	Red
50	Yellow
70	Orange
90	Blue
100	Dark Blue

Score	Color	Description
100	Dark Blue	to 90
90	Blue	to 70
70	Orange	to 50
50	Yellow	to 0
	Red	should not be interpreted, may be disordered

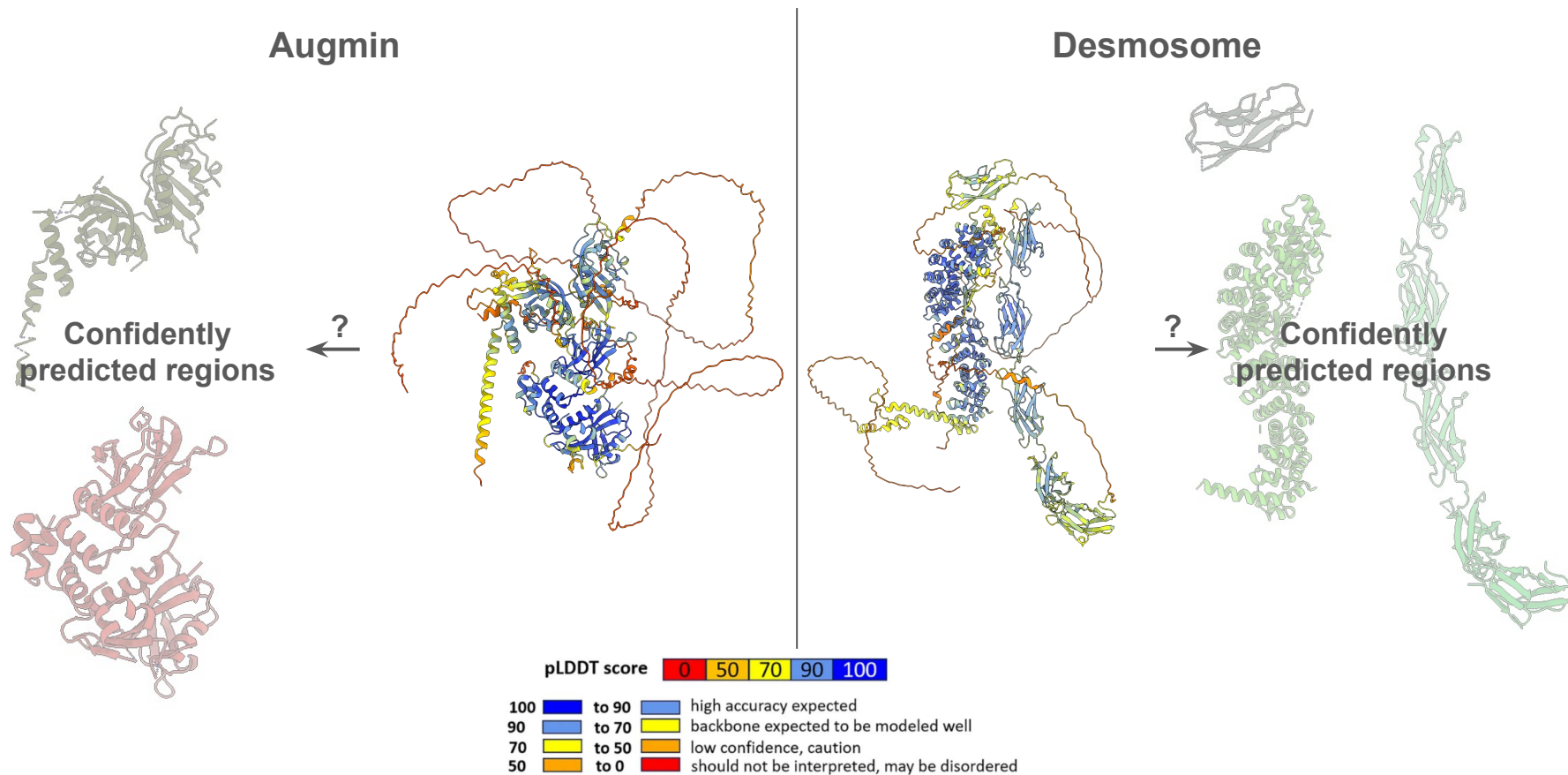
high accuracy expected
backbone expected to be modeled well
low confidence, caution



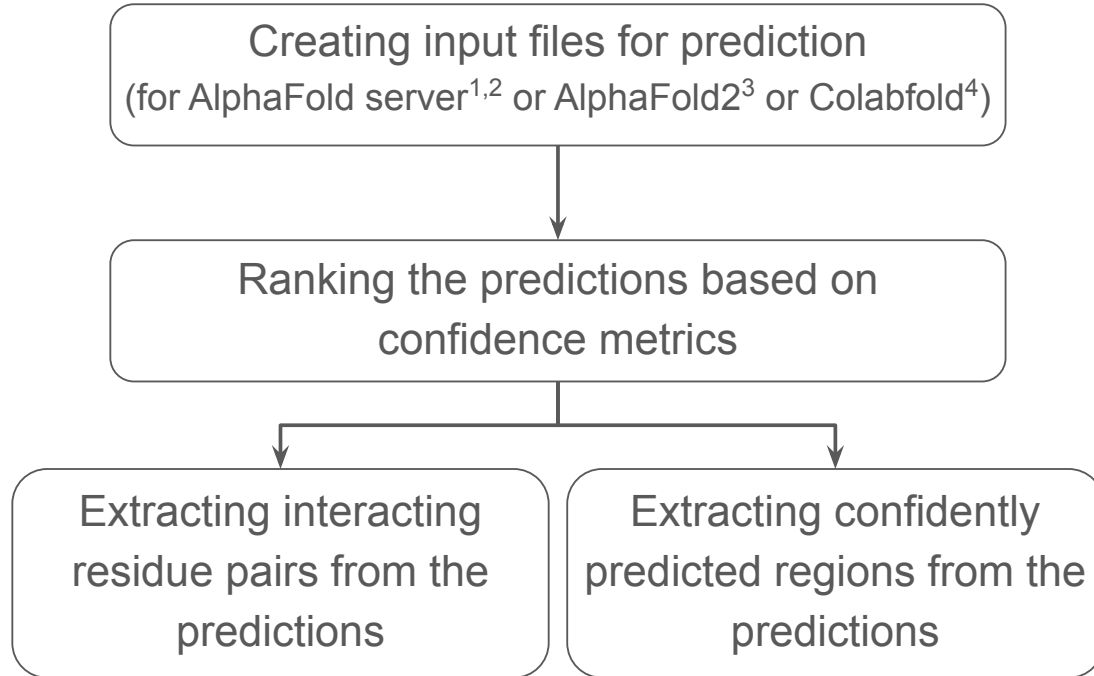
ipTM = 0.61
pTM = 0.46



Why do we need AF-Pipeline?



AF-Pipeline Workflow



1. AlphaFold Server. alphafoldserver.com. Available at: <https://alphafoldserver.com/>. ↩

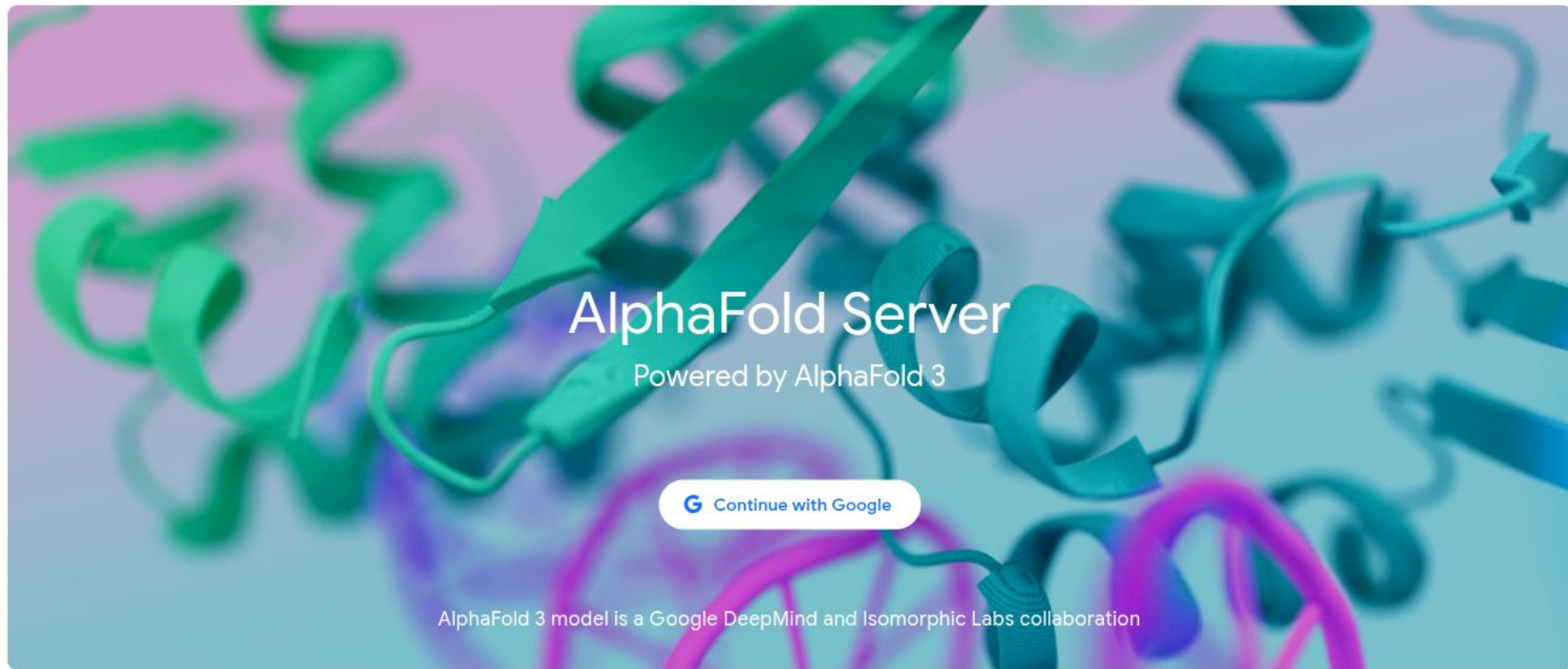
2. Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 630, 493–500 (2024). (<https://alphafoldserver.com/>) ↩

3. Jumper, J. et al. Highly Accurate Protein Structure Prediction with AlphaFold. Nature 596, 583–589 (2021). ↩

4. Mirdita, M. et al. ColabFold: making protein folding accessible to all. Nature Methods 19, 679–682 (2022).

(<https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb>) ↩

AlphaFold server



What can we model using AlphaFold?

Entity count

Additional options

Entity type

Protein

Copies

1

Protein ✓

DNA

RNA

Ligand

Ion

Allowed entities

AMINO ACID SEQUENCE OR
NUCLEIC ACID SEQUENCE OR
LIGAND CCD CODE OR
ION

PTMs

Custom MSA

Template settings

Delete

MDDDDIAALVV	DNGSGMCKAG	FAGDDAPRAV	FPSIVGRPRH	QGVVMVGMGQK	DSYVGDEAQS
70	80	90	100	110	120
KRGILTLKYP	IEHGIVTNWD	DMEKIWHHTF	YNELRVAPEE	HPVLLTEAPL	NPKANREKMT
130	140	150	160	170	180
QIMFETFNTP	AMYVAIQAVL	SLYASGRTTG	IVMDSGDGVV	HTVPIYEGYA	LPHAILRLDL
190	200	210	220	230	240
AGRDLTDYLM	KILTERGYSF	TTTAEREIVR	DIKEKLCYVA	LDFEQEMATA	ASSSSLEKSY
250	260	270	280	290	300
ELPDGQVITI	GNERFRCPEA	LFQPSFLGME	SCGIHETTFN	SIMKCDVDIR	KDLYANTVLS
310	320	330	340	350	360
GGTTMYPGIA	DRMQKEITAL	APSTMKIKII	APPERKYSVW	IGGSILASLS	TFQQMWISKQ
370	375				
EYDESGPSIV	HRKCF				

```
[
  {
    "name": "actin_monomer",
    "modelSeeds": [
      1
    ],
    "sequences": [
      {
        "proteinChain": {
          "sequence":
            "MDDIAALVVDNGSGMCKAGFAGDDAPRAVFPSIVGRPRHQGVMMGMGQKDSYVGDEAQ
            SKRGILTLKYPIEHGIVTNWDDMEKIWHHTFYNELRVAPEEHPVLLTEAPLNPKANREKM
            TQIMFETFNTPAMYVAIQAVLSLYASGRTTGIVMDSGDGVTHTVPIYEGYALPHAILRLD
            LAGRDLDYLMKILTERGYSFTTTAEREIVRDIKEKLCYVALDFEQEMATAASSSSLEKS
            YELPDGQVITIGNERFRCPEALFQPSFLGMESCGIHETTFNSIMKCDVDIRKDLYANTVL
            SGGTTMYPGIADRMQKEITALAPSTMKIKIIAPPERKYSVWIGGSILASLSTFQQMWISK
            QEYDESGPSIVHRKCF",
          "glycans": [],
          "modifications": [],
          "count": 1,
          "useStructureTemplate": true,
          "maxTemplateDate": "2021-09-30"
        }
      }
    ]
  }
]
```

AF-Pipeline config

```
protein_uniprot_map:
  Act1: "P60709"

af_input_jobs:
  - modelSeeds: [1]
    entities:
      - name: "Act1"
        type: "proteinChain"
```



https://github.com/isblab/af_pipeline/blob/main/examples/input/config.yaml


```
[
  {
    "name": "actin_monomer",
    "modelSeeds": [
      1
    ],
    "sequences": [
      {
        "proteinChain": {
          "sequence":
            "MDDIAALVVDNGSGMCKAGFAGDDAPRAVFPSIVGRPRHQGVMVGMGQKDSYVGDEAQ
            SKRGILTLKYPIEHGIVTNWDDMEKIWHHTFYNELRVAPEEHPVLLTEAPLNPKANREKM
            TQIMFETNTPAMYVAIQAVLSLYASGRITGIVMDSGDGVTHTVPIYEGYALPHAHLRLD
            LAGRDLTDYLMKILTERGYSFTTTAEREIVRDIKEKLCYVALDFEQEMATAASSSSLEKS
            YELPDGQVITIGNERFRCPEALFQPSFLGMESCGIHETTFNSIMKCDVDIRKDLYANTVL
            SGGTTMYPGIADRMQKEITALAPSTMKIKIIAPPERKYSVWIGGSILASLSTFQQMWISK
            QEYDESGPSIVHRKCF",
          "glycans": [],
          "modifications": [],
          "count": 1,
          "useStructureTemplate": true,
          "maxTemplateDate": "2021-09-30"
        }
      }
    ]
  }
]
```

AF-Pipeline config

```
protein_uniprot_map:
  Act1: "P60709"

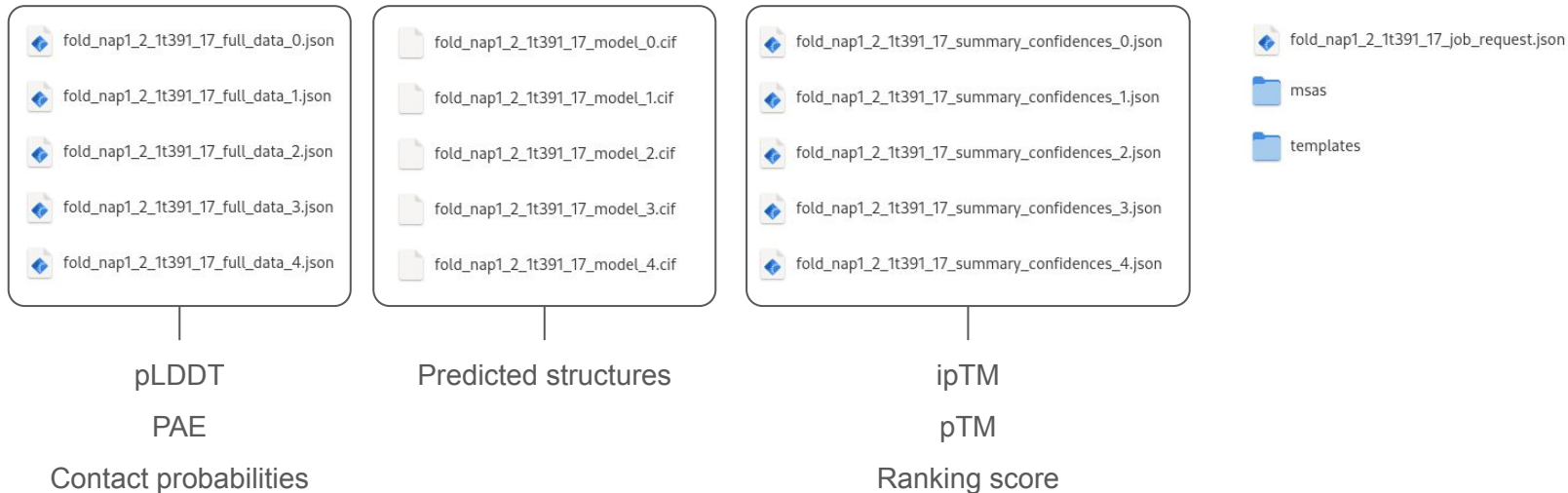
af_input_jobs:
  - modelSeeds: [1]
    entities:
      - name: "Act1"
        type: "proteinChain"
```



Open in Colab

https://github.com/isblab/af_pipeline

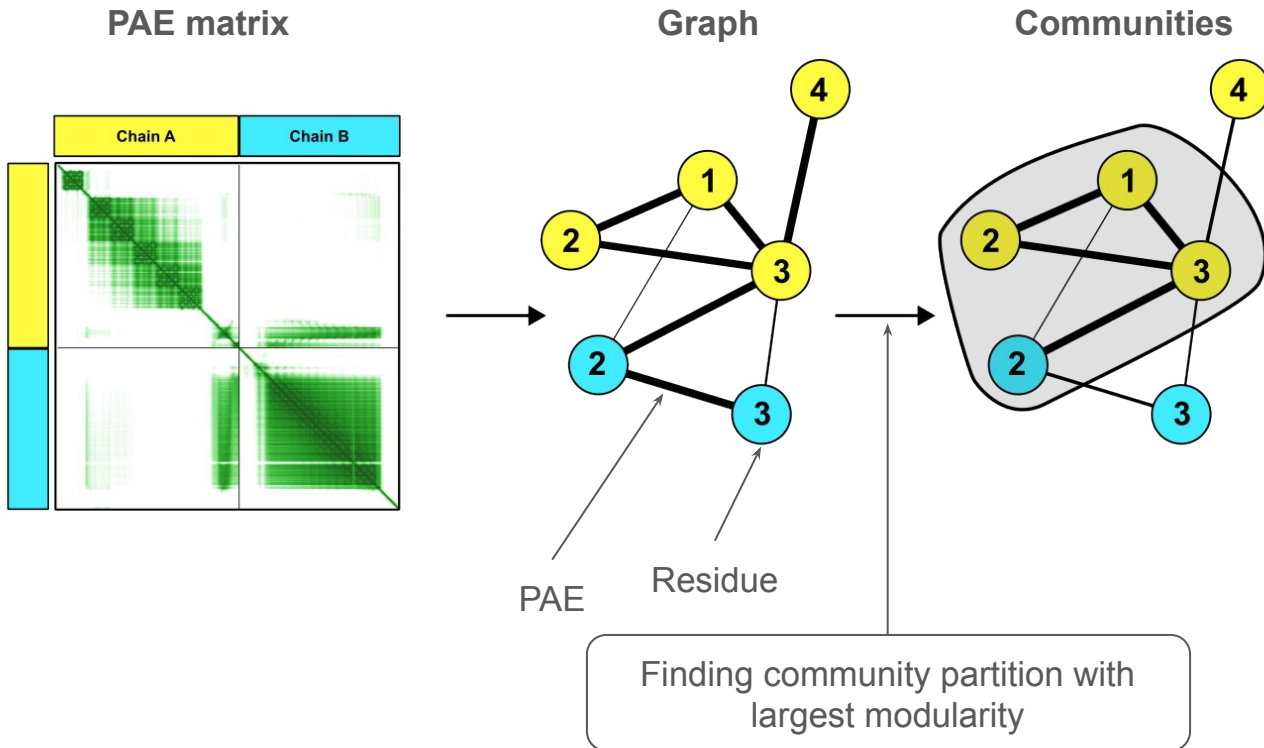
AlphaFold server prediction output

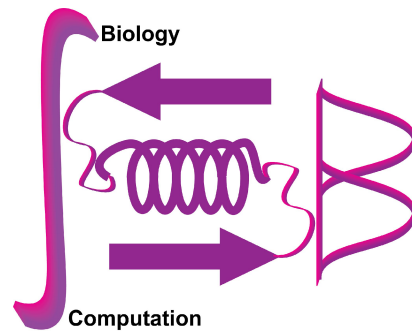


Identifying confident residue-pair interactions

- Distance between the representative atoms $\leq 8 \text{ \AA}$
- **Representative atoms:**
 - Proteins: $C\beta$ or $C\alpha$ (in case $C\beta$ is absent)
 - DNA or RNA: $C4$ in case of purines and $C2$ in case of pyrimidine
 - Ion
 - Ligand ??

Extracting confidently predicted regions





THANK YOU

Presenters:

Kartik Majila

Omkar Golatkar